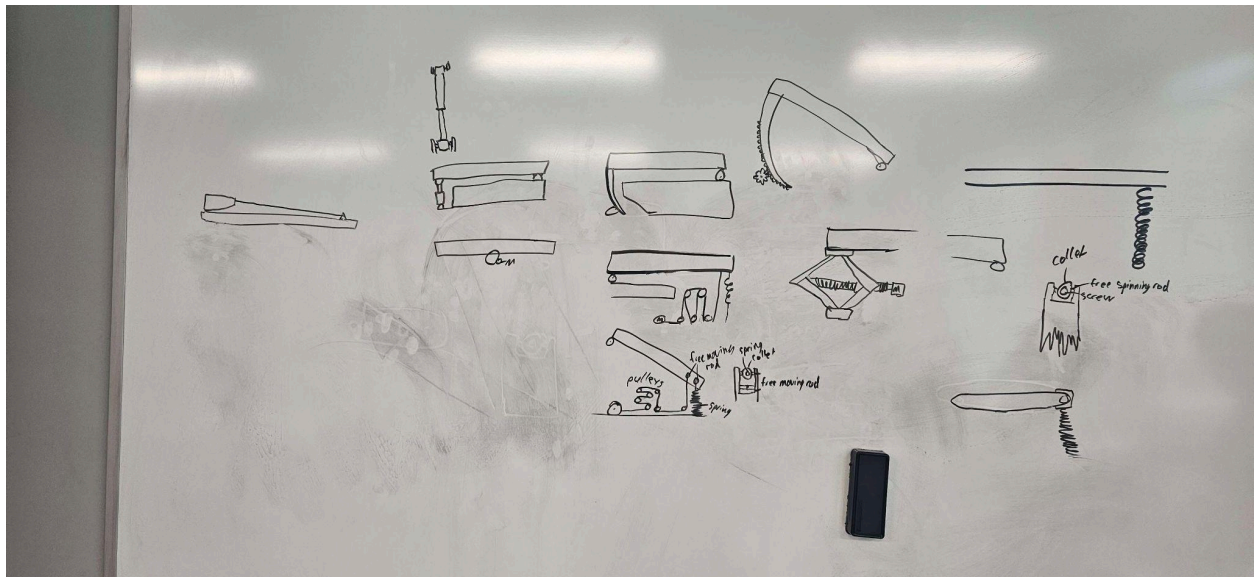


Element D - Jay

For the physical aspect of the automation, we separated our ideas into two categories: the elevation mechanism, and azimuth mechanism.

For the elevation mechanism, we began by brainstorming 6 different designs for automation of the feeder.

**Ideas 1, 2, 4, 5, and 6 all would have one end of the elevating platform placed on a hinge, elevated above the base. As such, only the pitch's mechanism will be described in each idea, while the hinge will be described only once in idea 1.*



1. Hydraulic Lever

- One end of the elevating platform would be on a hinge, while the other would be connected to a hydraulic piston. Once the piston extended, one end of the platform would be lifted up, increasing the pitch. The elevating platform could also be placed higher than the base of the system; instead of raising one end up, the extended piston would be level with the base, and retracting the piston would increase the pitch instead.

2. Motorized Lever

- The elevating platform would be attached to the base in the center by a rod. This rod would be connected to a motor. To reduce torque on the motor, the connection between the motor and platform would either be a belt and gear system, or simply gears spinning together .

3. Rack and Pinion

- The mechanized end would be on a section of ring gear, connected to a gear and motor. Once the motor spun, rotating the ring gear would cause the base to elevate.

Ideas 1, 2, 4, 5, and 6 all would have one end of the elevating platform placed on a hinge, elevated above the base. As such

4. Pulley

- The mechanized end would have two components: a spring, and a string. The spring would be in between the elevating platform and base, keeping the elevating platform level. The string would be strung through a few pulleys, and eventually into a spool on a motor. For space issues, the pulley would have to be below the elevated platform, placed atop the base. The motor would spin, drawing the string and subsequently lowering one end of the elevating platform, increasing pitch. The motor would spin in the opposite direction, allowing the spring to level the elevating platform.

5. Bar Lift

- The mechanized end would have a bar lift sandwiched between the elevating platform and base. Most likely based off a car jack, the center adjustable screw

would be spun to either extend or retract the bar lift. This extension/retraction would be performed by a motor connected to the adjustable screw.

6. Adjustable Screw

- The mechanized end would be similar to the pulley, but instead of string, the elevating platform would be attached to an adjustable screw, which, once spun, would allow the elevating platform to raise up and down. To compensate for the changing of the angle, the adjustable screw would have to be connected to the elevating platform with a moving slot

After creating a rough design for each concept, we made a decision matrix to decide which concept to go with. We based the criteria for the matrix off of:

- Simplicity - How complex were the parts this would be made out of?
- Ease of Assembly - How easy would it be for us, and subsequently the zookeepers, to assemble the product?
- Weight - Would the assembled product be unwieldy due to its weight?
- Cost - Would the cost of materials be feasible for us to procure?
- Size - Would the assembled product be unwieldy by size alone?

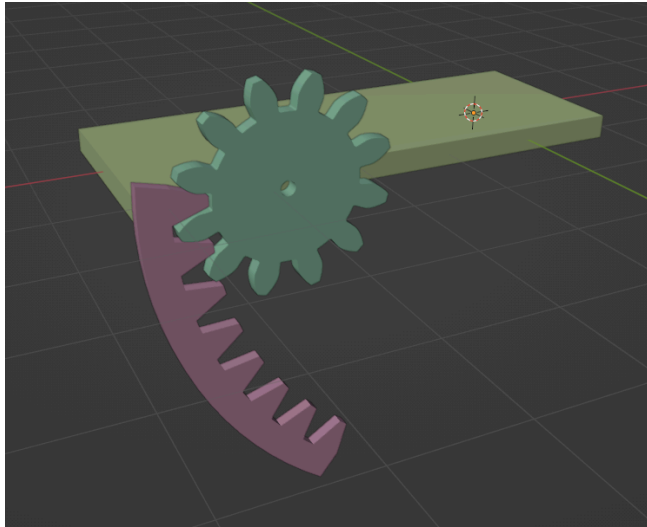
Each of us filled in the boxes by ourselves for each, then added the results up to a chart with everybody's input combined (shown below).

Criteria (5 best, 1 worst)	Hydraulic	Lever	Rack and Pinion	Pulley	Bar Lift	Adjustable Screw
Simplicity	5	6	13	5	14	11

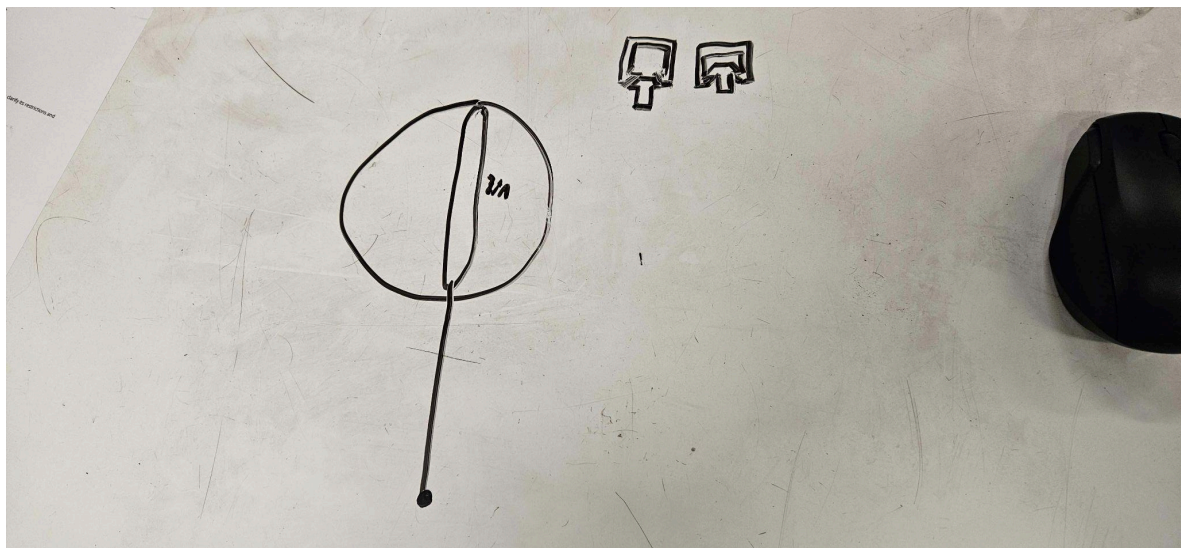
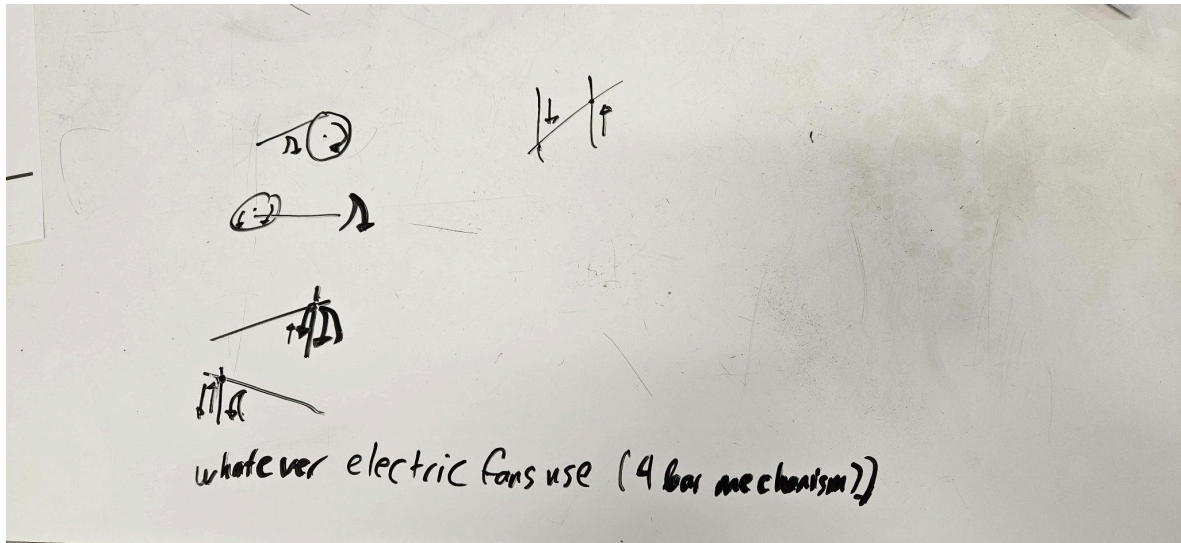
Ease of Assembly	5	8	10	4	11	11
Weight	7	11	11	7	7	12
Cost	4	12	9	6	6	8
Size	5	9	12	8	5	12
Total	26	46	55	30	43	54

Two options stood out as the most popular: the Rack and Pinion (shortened to RnP), and the Adjustable Screw (shortened to AS). The RnP idea had the second highest simplicity rating out of everything total, and its weight was also considered to be very low. It was also perceived to be one of the cheaper ideas, while also being very small. The AS idea had a simplicity rating a little below RnP, mainly due to concerns relating to how to rotate the adjustable screw *and* cause that motion to allow the elevating platform to rise/fall. It was considered the easiest to assemble, however, due to the seemingly hardest component to create was a cutout for the adjustable screw to slide on. Due to the elevation mechanism being only a screw, motor, and a few fittings, the weight and size seemed to be the lightest and smallest out of any option we had. Both seemed decent options, but some further thought was put into the AS design, and trying to get the physical screw's rotational motion to smoothly translate to vertical motion seemed more difficult than it was worth;

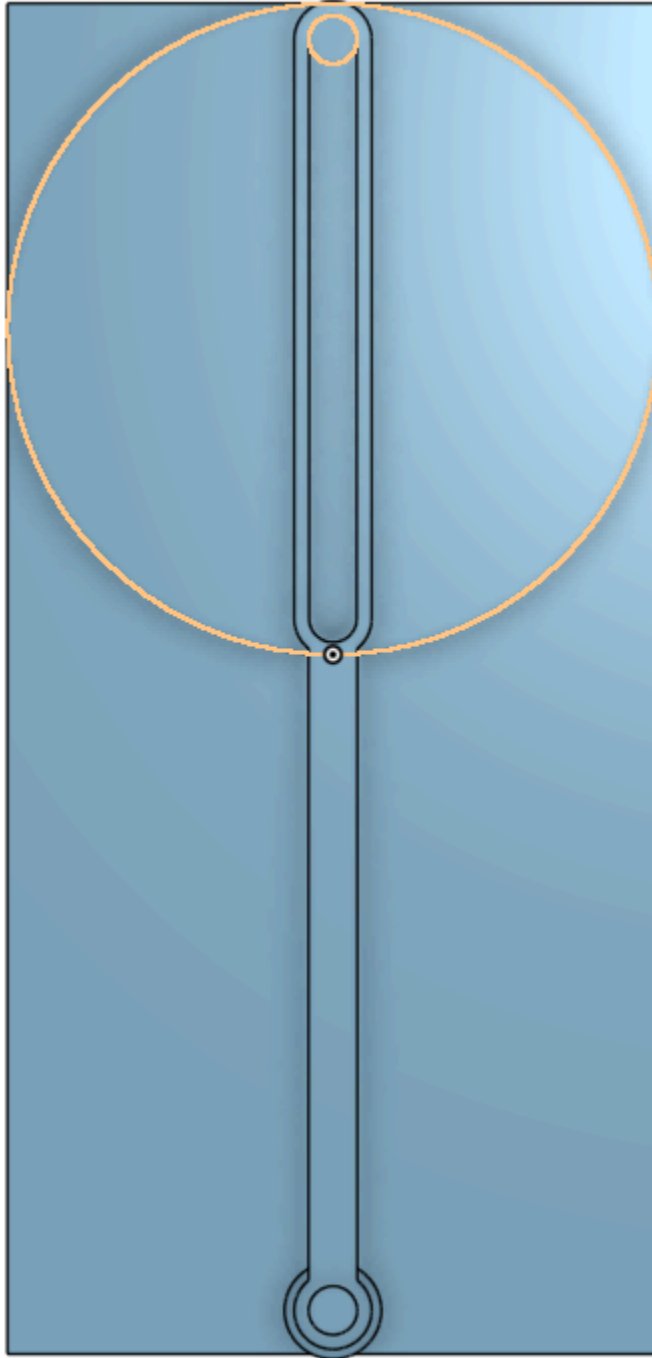
The rough sketch we had developed ended up as a concept similar to this.



For the horizontal mechanism, only 3 ideas were developed

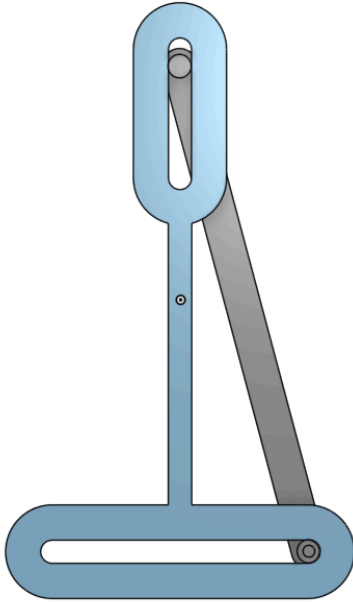


1. "Piston" - The idea behind a regular piston would be used, except the connecting rod of a regular piston would be where the base of the elevating platform would be placed atop of.



2. Twist - Two pins would be in a vertical and horizontal rail, and the elevating platform would be placed atop a beam connecting the two pins. The movement between the

horizontal pin would be mechanized, and the location of it would be used with trigonometry to determine the rotation of the system.



The image is from the bottom up of the system, to show the slots the pins would follow.

3. Gears - Simply using two different sized gears, one of which would rotate with the base.